



PROJECT REPORT

CREDIT CARD APPROVAL PREDICTION USING MACHINE LEARNING

Submitted By:

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1. INTRODUCTION

1.1 Project Overview

This project is a Machine Learning-based web application that predicts whether a credit card application will be approved or rejected based on applicant data. It uses historical financial data and machine learning algorithms to make predictions.

Credit cards are one of the most widely used financial services in the banking sector.

Banks receive thousands of credit card applications from customers every day.

Approving applications manually is time-consuming and may lead to errors. Therefore, machine learning can be used to automate the credit card approval process and improve decision-making.

The Credit Card Approval Prediction System is a machine learning project that predicts whether a credit card application is likely to be approved or rejected based on the applicant's information. The system analyzes important features such as income, education, employment status, family details, and other financial information to make accurate predictions.

In this project, data preprocessing techniques are applied to clean and prepare the dataset. Different machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, and XGBoost, are trained and evaluated. The best-performing model is then saved and integrated into a Flask web application, allowing users to enter applicant details and receive real-time credit card approval predictions through a simple and user-friendly interface.

This project provides practical experience in data analysis, feature engineering, machine learning model development, and web application deployment. It also

demonstrates how artificial intelligence can help financial institutions make faster, more consistent, and more accurate credit approval decisions.

1.2 Purpose

The main purpose of this project is to automate the credit card approval process using Machine Learning techniques, reducing manual effort and improving accuracy in decision-making.

The main purpose of this project is to develop a machine learning-based Credit Card Approval Prediction System that can automatically predict whether a credit card application should be approved or rejected based on the applicant's financial and personal information. The system aims to reduce manual effort, improve decision-making accuracy, and speed up the credit approval process. It also provides practical experience in data preprocessing, machine learning model development, model evaluation, and Flask web application deployment for real-time predictions.

2. IDEATION PHASE

2.1 Problem Statement

Banks face difficulty in manually verifying large numbers of credit card applications. This project solves this by predicting approval or rejection using machine learning models.

2.2 Empathy Map Canvas

- **Think & Feel:** Applicants want quick approval decisions
 - **Hear:** Bank policies and eligibility rules
 - **See:** Complex manual approval process
 - **Say & Do:** Submit applications and wait for results
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2.3 Brainstorming

- Use ML models for prediction
- Build web interface using Flask
- Improve accuracy using multiple algorithms
- Deploy model for real-time prediction

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

User enters data → System processes data → Model predicts → Result displayed

3.2 Solution Requirement

- Python programming
 - Machine Learning models
 - Flask web framework
 - HTML/CSS frontend
 - Dataset for training
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3.3 Data Flow Diagram

Data → Preprocessing → Model Training → Prediction → Output

3.4 Technology Stack

- Python
 - Flask
 - Pandas, NumPy
 - Scikit-learn
 - XGBoost
 - HTML, CSS
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4. PROJECT DESIGN

4.1 Problem Solution Fit

Machine Learning models are suitable for predicting credit approval based on historical data patterns.

4.2 Proposed Solution

A web-based system that takes user input and predicts credit card approval using a trained ML model.

4.3 Solution Architecture

Frontend (HTML) → Flask Backend → ML Model → Prediction Output

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Data collection
 - Data preprocessing
 - Model training
 - Model evaluation
 - Web development
 - Deployment
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6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

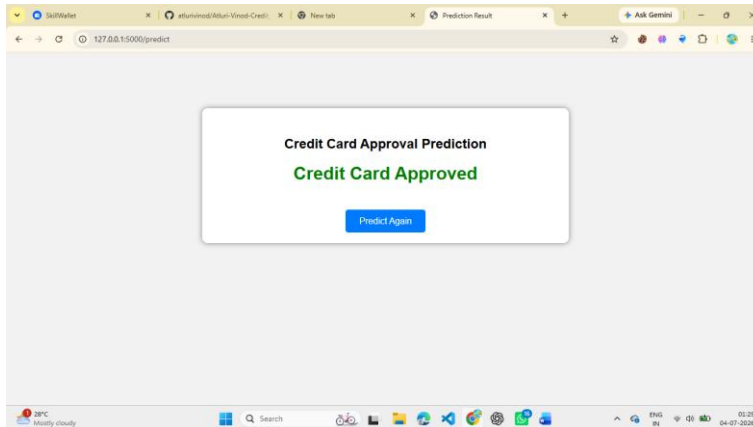
- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost

Best performance achieved using Random Forest Classifier.

7. RESULTS

7.1 Output Screenshots

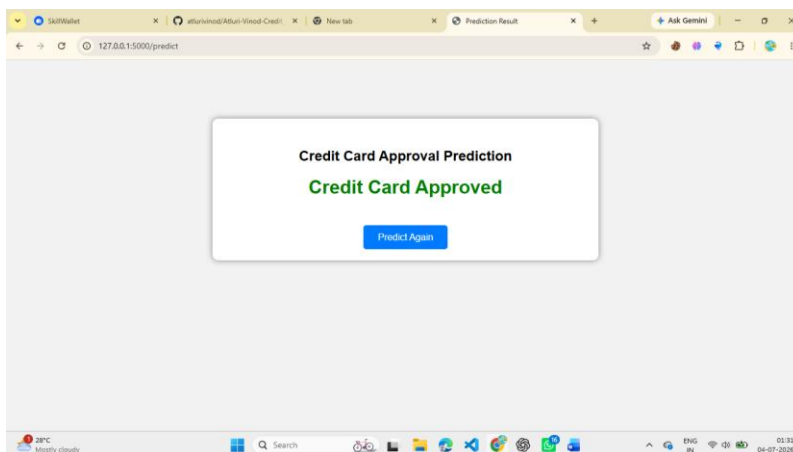
- Home Page



- Input Form Page

A screenshot of a web browser displaying the 'Credit Card Approval Prediction' input form page. The page has a light gray background. In the center, there is a white rectangular box with a thin gray border. Inside this box, the text 'Credit Card Approval Prediction' is at the top in a small, dark font. Below it, there is a list of input fields for user information: ID, Gender (Male=1 Female=0), Own Car (Yes=1 No=0), Own House (Yes=1 No=0), Number of Children, Annual Income, Income Type (Encoded), Education Type (Encoded), Family Status (Encoded), Housing Type (Encoded), and Date of Birth. Each field is a simple text input box. The browser's address bar shows the URL '127.0.0.1:5000/predict_page'. The Windows taskbar is visible at the bottom of the screen.

- Prediction Result Page



8. ADVANTAGES & DISADVANTAGES

Advantages

- Fast decision making
- Reduces manual effort
- High accuracy prediction

Disadvantages

- Depends on training data quality
 - Needs proper feature engineering
 - Not 100% error-free
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9. CONCLUSION

This project successfully demonstrates how Machine Learning can be used to automate credit card approval prediction with good accuracy and real-time web application integration.

10. FUTURE SCOPE

- Deploy on cloud (AWS / Render)
 - Improve accuracy using Deep Learning
 - Add user authentication system
 - Real-time bank integration
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11. APPENDIX

Source Code

Available in GitHub repository

Dataset Link

https://drive.google.com/file/d/13Yr0asj8gXFI1oQTFWWo557wJlNa9s2x/view?usp=drive_link

GitHub & Project Demo Link

https://github.com/atlurivinod/Atluri-Vinod-Credit_Card_Approval_Prediction
